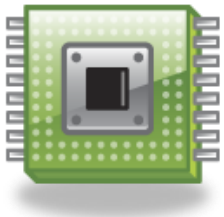

Homework #1 (Deadline : 2025/10/14 Tue. 23:59)

- Design a combinational 14-bit adder/subtractor by writing gate-level codes. The basic I/O specification is described below. Please use this spec. to design your circuit.
  - Two 14-bit binary inputs: A, B
  - One 1-bit control input for addition/subtraction: M (M=1 (0) for addition (subtraction))
  - One 14-bit sum output: sum
  - One 1-bit carry-out output: c_out
- Please specify some reasonable delays in your components and generate some waveforms to verify your design.
- Also design a testbench to verify your designs. Must include $finish command in your testbench. Describe what test inputs you use in the testbench to verify your design.
- Describe the longest path in your design and accordingly calculate or estimate the maximum delay in your design.
- Describe how many logic gates are used to implement your design.
- You are encouraged to implement different architectures (e.g., carry lookahead adder) and make comparisons.